

Dr. PAPA RAO CHALLAGUNDLA

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Objective

To work in the Electronics and Communication Engineering domain in a challenging academic and research environment where my 15 years of teaching, research, and administrative expertise are best utilized, contributing to institutional growth, curriculum development, advanced research in wireless communications and VLSI, and the holistic development of students.

Professional Experience

Associate Professor & Head of Department – ECE

GVR&S College of Engineering and Technology, Budampadu, Guntur

June 2023 – Present

Guntur, A.P., India

- Serving as Head of the ECE Department, overseeing academic planning, faculty management, timetable scheduling, and departmental administration for both UG and PG programs.
- Spearheading NAAC, NBA, and IQAC accreditation and compliance activities, ensuring the department consistently meets national quality benchmarks.
- Delivering advanced UG and PG courses including VLSI Technology, Embedded Systems, Wireless Communications, and Digital System Design through HDL.
- Organized and conducted Faculty Development Programs (FDPs) on IoT, PCB Design, MATLAB, and Embedded Systems to enhance faculty competency.
- Managing Memoranda of Understanding (MoUs) with industry and academic institutions to strengthen industry-institute interaction.
- Serving as In-Charge of Student Internship Programs, coordinating placement and skill development initiatives.
- Active member of the Research & Development Cell, Skill Development Centre, and Entrepreneur & Development Cell.
- Functioning as IETE Convener and Industry Institute Interaction Cell Co-Convener to bridge academic learning with industry requirements.

Associate Professor – Electronics & Communication Engineering

Guntur Engineering College, Yanamadala, Guntur

June 2017 – May 2023

Guntur, A.P., India

- Taught a wide range of UG and PG subjects including VLSI Technology & Design, Low Power VLSI, CPLD & FPGA Architecture, Wireless Communications, Embedded Systems, and Digital Communications over six academic years.
- Conducted a 6-Day Faculty Development Program on “Internet of Things (IoT)” and a 3-Day FDP on “PCB Design” for faculty members at GEC, Guntur.
- Conducted a 3-Day Workshop on “Embedded Systems” and a 3-Day Workshop on “PCB Design” for students to enhance practical skills.
- Attended multiple AICTE-sponsored, APSSDC, and NPTEL Faculty Development Programs to remain current with emerging technologies.
- Guided multiple UG student projects in the areas of embedded systems, wireless communication, and IoT-based applications.
- Participated as a member in institutional quality assurance and accreditation activities.

Associate Professor – Electronics & Communication Engineering

Tirumala Engineering College, Jonnalagadda, Narasaraopet

December 2015 – May 2017

Guntur, A.P., India

- Delivered UG courses in Digital Logic Design, Microprocessors & Microcontrollers, Analog & Digital Communications, and Wireless Communications.
- Conducted a 2-Day FDP on “MATLAB and Simulink” and a 2-Day FDP on “ASIC Design Flow Using Mentor Graphics” for faculty members.
- Organized workshops on “Web Entrepreneurship” and “Employer & Employability” to prepare students for industry readiness.
- Attended a 3-Day FDP on “Excerpts of Research Methodology and Processes” to strengthen research aptitude.

Assistant Professor – Electronics & Communication Engineering

Tirumala Engineering College, Jonnalagadda, Narasaraopet

June 2015 – November 2015

Guntur, A.P., India

- Taught core ECE subjects at UG level including Switching Theory & Logic Design, IC Applications, and Computer Networks.
- Supported departmental academic and co-curricular activities as part of the ECE faculty team.

Assistant Professor – Electronics & Communication Engineering

St. Peter's Engineering College, Dhulapally

September 2014 – June 2015

Hyderabad, Telangana, India

- Delivered UG-level courses including Embedded Systems, Digital Communications, and Computer Architecture & Organization.
- Guided student mini-projects in embedded and communication domains; contributed to departmental examination and evaluation processes.

Assistant Professor – Electronics & Communication Engineering

Noble College of Engineering & Technology for Women, Nadargul

July 2010 – August 2014

Hyderabad, Telangana, India

- Taught UG courses including Digital Logic Design, Microprocessors, Cellular & Mobile Communications, and Telecommunication Switching Systems over four years.
- Conducted a Workshop on “Ethical Hacking” to expose students to cybersecurity concepts and build practical awareness.
- Actively participated in academic planning, internal examinations, and student mentoring activities.

Academic Qualifications

Doctor of Philosophy (Ph.D.) – Electronics & Communication Engineering

Jawaharlal Nehru Technological University Hyderabad (JNTUH)

2025

Hyderabad, India

- Thesis Title: “Performance Improvement with Load Balancing using Adaptive Filters for Wireless Communication”
- Research focused on adaptive filter-based techniques for load balancing, BER optimization, and signal detection in MIMO and CDMA wireless systems.

Master of Technology (M.Tech.) – VLSI System Design

Jawaharlal Nehru Technological University Hyderabad (JNTUH)

2010

Hyderabad, India

- Specialized in VLSI design methodologies, CMOS circuit design, FPGA-based digital systems, and low-power VLSI techniques.

Bachelor of Technology (B.Tech.) – Electronics & Communication Engineering

Jawaharlal Nehru Technological University Hyderabad (JNTUH)

2008

Hyderabad, India

- Core studies in analog/digital electronics, signal processing, microprocessors, communication systems, and antenna theory.

Diploma – Electronics & Communication Engineering

State Board of Technical Education & Training (SBTE&T), Andhra Pradesh

2005

Andhra Pradesh, India

Secondary School Certificate (S.S.C.)

Board of Secondary Education (B.S.E.), Andhra Pradesh

1998

Andhra Pradesh, India

Research

Ph.D. Research Thesis: “Performance Improvement with Load Balancing using Adaptive Filters for Wireless Communication”

The research investigates the application of adaptive filtering algorithms for optimizing load balancing in MIMO-based LTE and LTE-Advanced wireless networks. The work focuses on reducing Bit Error Rate (BER) at linear filter receivers, improving signal detection efficiency using multiuser detection techniques, and enhancing the spectral efficiency of CDMA and OFDM-based systems under fading channel conditions. Key contributions include the proposed Filter-Based Optimal Transmission Path Scheduling (FOTPS) algorithm and the Rule Induction Framework for Load Balancing (RIF-LB) validated through simulations.

Technical Skills

Operating Systems	MS-DOS, Windows 2007, Windows 2008/2010
EDA & Design Tools	Xilinx Vivado (FPGA Design), Mentor Graphics (ASIC/PCB Design), MASM, TASM
Simulation Tools	MATLAB, Simulink (System Modeling, Signal Processing, Communication Systems)
Embedded Platforms	Arduino (Microcontroller Programming, IoT Prototyping, Sensor Integration)
Core Research Areas	MIMO Systems, OFDM, CDMA/MCCDMA, Adaptive Filters, BER Optimization, Load Balancing
Areas of Interest	Wireless Communications, Embedded Systems, Signal Processing, Image Processing, Radar Systems
Teaching Domains	VLSI Design, Digital Electronics, Microprocessors, Telecommunication Systems, IoT, Computer Networks

Courses Taught

Undergraduate (UG) Level:

Digital Logic Design • DSD & Digital IC Applications • Switching Theory & Logic Design • Microprocessors & Microcontrollers • VLSI Technology • Wireless Communications & Networks • Telecommunication Switching Systems & Networks • Embedded Systems • Cellular & Mobile Communications • IC Applications • Analog Communications • Digital Communications • Computer Architecture & Organization • Computer Networks • IPR & Patents • Internet of Things (IoT) • Communication Standards & Protocols • Wireless Communications

Postgraduate (PG) Level:

Digital System Design • CPLD & FPGA Architecture • VLSI Technology & Design • Low Power VLSI Technology • Microcontroller for Embedded System Design • System Modeling & Simulation • Embedded System Design • Digital System Design Through HDL

Projects Guided / Done

- **Paralysis Patient Healthcare Monitoring System using Flex Sensor** – IoT-based health monitoring solution capturing limb movement data via flex sensors to assist paralysis patients in real-time clinical monitoring.
- **Keyless Door Locking System using Arduino** – Microcontroller-based secure access control system eliminating physical keys through digital authentication.
- **Mobile Signal Transition Detector** – Device designed to detect transitions in mobile network signals for communication fault diagnosis.
- **Airport Security Luggage Scanner System** – Embedded system solution simulating baggage screening and threat detection at airport security checkpoints.

- **Solar Highway Lighting System** – Renewable energy-based automated highway lighting system using solar panels and light-sensitive switching circuits.
- **RF Based Industrial Applications Control Systems** – Radio-frequency controlled industrial automation system enabling remote operation of industrial loads.
- **Advanced Railway Fault Detection Using Zigbee** – Wireless sensor network-based railway track fault detection system using Zigbee protocol for real-time alerts.
- **Automatic Car Parking System using IR Sensors** – IR sensor-based automated vehicle detection and parking management system reducing manual monitoring overhead.
- **Reduction of BER in Receivers for Wireless Communication using OFDM-MIMO** – Research-level simulation project implementing OFDM combined with MIMO antenna techniques to significantly reduce Bit Error Rate in wireless receivers.

Journal Publications – 11 Publications (SCOPUS, Web of Science, UGC Approved)

- [1] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Filter Based Optimal Transmission Path Scheduling (FOTPS) in MIMO LTE and LTE-A Networks**. *International Journal of Networking and Virtual Organisations (IJNVO)*, Inderscience Publishers. [SCOPUS Indexed], ISSN Online: 1741-5225, Vol.23, No.4, 2020, pp.390–402.
- [2] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Rule Induction Framework for Load Balancing (RIF-LB) for Optimal BER at Linear Filter Receivers of Self Organized MIMO Cellular Networks**. *Journal of Advanced Research in Dynamical and Control Systems (JARDCS)*. [SCOPUS Indexed], ISSN: 1943-023X, Vol.10, Special Issue 08, 2018, pp.838–853.
- [3] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Low Complexity Signal Detector for MIMO MCCDMA System for Longer Delay Channel**. [SCOPUS Indexed], ISSN: 2227-524X, Vol.7, No.3, 2018, pp.423–426.
- [4] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Variable Rate Spreading Scheme for MCCDMA System over Longer Delay Channels**. *Research Journal of Pharmaceutical, Biological and Chemical Sciences (RJPBCS)*. [Web of Science / UGC Approved], ISSN: 0975-8585, Vol.8, No.2, pp.634–638.
- [5] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY, Jagadeesh Thati. **Optimum Bit Error Rate for Linear Multiuser Detection in Wireless Communications**. *Elixir International Journal*. [UGC Approved], ISSN: 2229-712X.
- [6] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY, SRINIVASA RAO ALLURI. **Multiuser Detection using BPSK Modulation with Adaptive Filters**. *Elixir International Journal*. [UGC Approved], ISSN: 2229-712X.
- [7] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Performance of Adaptive Frequency Hopping in Group-Orthogonal Code-Division Multiplex**. *International Journal of Mechanical and Technology Engineering (IJMTE)*. [UGC Approved], ISSN: 2249-7455, Vol.8, Issue XII, December 2018, pp.921–927.
- [8] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Performance of Fading and Gaussian Noise in Direct-Sequence Code-Division Multiple-Access (DS-CDMA) Systems**. *Journal of Advanced Sciences and Computing (JASC)*. [UGC Approved], ISSN: 1076-5131, Vol.4, No.6, November 2017, pp.207–217.
- [9] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **An Implementation of Fully Integrated 24-GHz Multi-Band Phased-Array Transmitter in CMOS**. *International Journal of Scientific Engineering and Technology Research (IJSETR)*, Vol.02, Issue 15, November 2013, pp.1725–1732.
- [10] B. Ramya Sri, CH. PAPARAO. **Automatic Car Parking System using IR Sensors**. *International Journal of Engineering Science and Computing (IJESC)*, ISSN: 2321-3361, Vol.7, No.4, April 2017, pp.6485–6487.
- [11] GUNTUPALLI MANISRI, CHALLAGUNDLA PAPA RAO. **Design a High Speed Bypass Multiplier for Communication**. *International Journal of Research (IJR)*. [UGC Approved], eISSN: 2348-6848, pISSN: 2348-795X, Vol.05, Issue 12, April 2018.

Conference Publications

International Conferences:

- [1] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Performance and Power Efficiency of Complementary Codes Aided MIMO MCCDMA Transmission for Broadband Vehicular Networks**. *IEEE International Conference on Emerging Electronics, Communication and Computing Technologies (IEEE-EECCMC-2018)*. [IEEE Indexed].
- [2] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY. **Reduction of BER in Receivers for Wireless Communication Using MIMO OFDM**. *International Conference on Recent Innovations in Engineering and Applied Technology (ICRIEAT-2017)*.

National Conferences:

- [1] CH. PAPARAO, Dr.P. SUMITHA BHASHINI, Dr.P. CHANDRASEKHAR REDDY, P. MANOHARA RAO. **Perception of Complexity and Interference of User Transmitting Signal being Resolved by using OOTBC in MIMO**. *National Technical Symposium on Emerging Technologies (NTSET-2018)*.

Faculty Development Programs (FDPs) Conducted

- **6-Day FDP on “Internet of Things (IoT)”** – Organized and conducted for faculty members at Guntur Engineering College (GEC), Guntur, covering IoT architecture, protocols, sensor integration, and real-time applications.

- **3-Day FDP on “PCB Design”** – Conducted at GEC, Guntur, covering PCB layout design, routing techniques, and fabrication workflow using industry-standard tools.
- **2-Day FDP on “MATLAB and Simulink”** – Conducted at Tirumala Engineering College (TEC), Narasaraopet, covering signal processing, system modeling, and simulation for engineering applications.
- **2-Day FDP on “ASIC Design Flow Using Mentor Graphics”** – Conducted at TEC, Narasaraopet, providing hands-on training in RTL design, synthesis, simulation, and layout using Mentor Graphics EDA suite.

Faculty Development Programs (FDPs) Attended

- **12-Week NPTEL Online Course on “Cloud Computing”** – Completed with certificate; covered cloud architecture, virtualization, deployment models (IaaS, PaaS, SaaS), and cloud security fundamentals.
- **12-Week NPTEL Online Course on “Computer Networks and Internet Protocols”** – Completed with certificate; covered TCP/IP stack, routing protocols, network security, and internet architecture.
- **1-Week AICTE FDP on “Amazon Web Services (AWS)”** – National Engineering College (NEC), in collaboration with AICTE; covered AWS cloud services, EC2, S3, and cloud deployment.
- **1-Week AICTE FDP on “Emerging Technologies in Wireless Communications”** – NEC, in collaboration with AICTE; explored 5G, mmWave, massive MIMO, and next-generation wireless systems.
- **1-Week AICTE FDP on “Cloud Infrastructure (AWS)”** – GEC, Guntur, in collaboration with AICTE; focused on cloud infrastructure design and management.
- **2-Week APSSDC FDP on “Machine Learning with Python Programming”** – GEC, Guntur, in collaboration with APSSDC; covered supervised/unsupervised learning, scikit-learn, and ML project development.
- **1-Week FDP on “Outcome Based Education (OBE)”** – KHIT, Chowdavaram; focused on OBE framework, CO-PO mapping, and curriculum design aligned to NBA/NAAC norms.
- **1-Week FDP on “Design Thinking for Innovation”** – Conducted by IQAC, PACE Ongole; covered human-centered design, ideation, prototyping, and innovation methodologies.
- **1-Week FDP on “Advanced Technologies & Challenges in Electronics and Communication Engineering”** – GATESIT, Ananthapuramu; explored current trends in VLSI, IoT, and communication technologies.
- **1-Week FDP on “Modern Research Trends in Communication, Signal Processing and VLSI/MEMS”** – ALIET, Vijayawada; covered emerging research directions in wireless, MEMS devices, and signal processing algorithms.
- **1-Week FDP on “Overview of AI & Its Applications in Speech Processing”** – CEC, Chebrolu; introduced deep learning models for speech recognition, NLP, and audio signal analysis.
- **1-Week FDP on “Machine Learning for Signal Processing and VLSI”** – Anurag University, Hyderabad; covered ML-based design automation for VLSI and ML algorithms for signal classification.
- **1-Week FDP on “Advanced Embedded System Design and Its Applications”** – SCET, Hyderabad; hands-on training in advanced microcontroller programming, RTOS, and embedded system prototyping.
- **1-Week FDP on “Online Teaching-Learning Process using ICT Tools for Education 4.0”** – St. Vincent Pallotti College of Engineering & Technology, Nagpur; covered LMS platforms, e-learning tools, and digital pedagogy.
- **1-Week FDP on “Design of Solar PV System using PVSYST Software”** – Vaagdevi College of Engineering, Hyderabad; practical training in solar panel sizing, simulation, and energy yield analysis using PVSYST.
- **1-Week FDP on “Moodle – Learning Management System”** – Vaagdevi Engineering College, Warangal; training in Moodle LMS setup, course creation, assessment design, and student management.
- **1-Week FDP on “AWS Cloud Computing”** – APSSDC, Vijayawada; covered AWS core services, serverless computing, and cloud deployment pipelines.
- **1-Week FDP on “Technological Solutions of Covid and Challenges in Society”** – BIET, Hyderabad; explored technology-driven solutions in healthcare, remote work, and society during the pandemic.
- **1-Week FDP on “Digital System Design Using HDL”** – CDAC, Hyderabad; covered VHDL/Verilog coding, synthesis, simulation, and FPGA implementation methodologies.
- **3-Day FDP on “VLSI Digital Circuits & Testing Techniques”** – MESI, Bhogapuram, Vijayanagaram; focused on digital VLSI design verification and fault testing approaches.
- **3-Day FDP on “Recent Advancements in Image Processing & IoT”** – LIMT, Madalavarigudem, Vijayawada; covered image segmentation, feature extraction, and IoT-integrated image processing systems.
- **3-Day FDP on “Excerpts of Research Methodology and Processes”** – TEC, Narasaraopet; covered research design, literature review, data analysis, and scholarly publication processes.

Workshops Conducted

- **Workshop on “Embedded Systems”** – GEC, Guntur: Comprehensive hands-on workshop covering microcontroller architecture, GPIO interfacing, serial communication, and real-time embedded programming.
- **Workshop on “PCB Design”** – GEC, Guntur: Practical workshop on schematic capture, PCB layout, component placement, signal routing, and design rule checks (DRC) using EDA tools.
- **Workshop on “Ethical Hacking”** – NETW, Nadergul: Student-focused workshop introducing ethical hacking concepts, network scanning, vulnerability exploitation, and cybersecurity best practices.
- **Workshop on “Web Entrepreneurship”** – TEC, Narasaraopet: Workshop on web-based business models, startup ecosystem, digital marketing, and entrepreneurship skills for engineering students.
- **Workshop on “Employer & Employability”** – TEC, Narasaraopet: Career development workshop covering resume building, aptitude training, communication skills, and interview preparation.

Workshops Attended

- **ATAL Workshop on “Emerging Research Area in Semiconductor Chip Design using AI-ML Algorithms for Neural Network Generation (NNGen) in Telugu Language”** – PACE Institute of Technology and Sciences, Ongole; covered AI-ML driven chip design automation and neural network hardware implementation trends.
- **Workshop on “IPR & Patents”** – Malineni Perumallu Educational Society, Guntur; covered intellectual property rights, patent filing procedures, copyright law, and technology commercialization.
- **Workshop on “X86 Microprocessors and Multi-Core Systems”** – JNTUK, Kakinada; covered x86 processor architecture, multi-core parallelism, cache coherence, and advanced assembly language programming.

Seminars / Webinars Attended

- **3-Day Webinar Series on “VLSI Design”** – ALIET, Vijayawada; covered advanced topics in CMOS VLSI design, timing analysis, power optimization, and physical design methodologies.
- **1-Day Webinar on “Industrial Embedded & IoT Applications”** – GEC, Guntur; explored real-world deployment of embedded systems and IoT platforms in industrial automation, smart manufacturing, and remote monitoring.

Administrative & Institutional Responsibilities

- **NAAC Coordinator** – Led preparation of Self Study Reports (SSR), data collection, documentation, and coordination with NAAC peer teams for institutional accreditation assessment.
- **NBA Coordinator** – Managed program-level accreditation documentation, CO-PO attainment calculations, and continuous quality improvement activities as per NBA criteria.
- **IQAC Coordinator** – Coordinated Internal Quality Assurance Cell activities including academic audits, feedback analysis, best practices documentation, and quality enhancement initiatives.
- **IETE Convener** – Organized technical events, seminars, and student chapter activities under the Institution of Electronics and Telecommunication Engineers (IETE).
- **Industry Institute Interaction Cell Co-Convener** – Facilitated industry visits, guest lectures, MoU signings, and collaborative projects to strengthen industry-academic partnerships.
- **In-Charge, Student Internship Programs** – Coordinated student internship placements with industry partners, tracked student progress, and ensured compliance with university internship requirements.
- **Member, Research & Development Cell** – Contributed to research proposal preparation, funding applications, and departmental R&D policy formulation.
- **Member, Skill Development Centre** – Supported skill-based training programs, certifications, and workshops to improve student employability.
- **Member, Entrepreneur & Development Cell** – Encouraged student startup initiatives, innovation projects, and entrepreneurship awareness programs.
- **In-Charge, MoUs** – Managed institutional MoUs with industry, academic bodies, and research organizations for collaborative academic and research activities.

NPTEL Certifications

- **Cloud Computing (12 Weeks)** – Completed NPTEL online course with certificate; topics covered include cloud service models, virtualization, cloud storage, security, and cloud deployment strategies.
- **Computer Networks and Internet Protocols (12 Weeks)** – Completed NPTEL online course with certificate; topics covered include OSI/TCP-IP model, routing algorithms, transport protocols, and network security fundamentals.

Professional Memberships

International Association of Engineers (IAENG) – 164280
Society for Digital Information and Wireless Communications (SDIWC) – 8976
International Association of Electrical, Electronics and Energy Engineers (IAEEEE) – 71050711

International Society for Automation and Control (ISAC) – RM2024013001
Computer Science Teachers Association (CSTA) – 198358027065

Personal Profile

Full Name: PAPA RAO CHALLAGUNDLA
Date of Birth: 17th November, 1983
Marital Status: Married

Father's Name: SAIDULU
Gender: Male
Languages Known: Telugu, English, Hindi

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief. I take full responsibility for the accuracy of the details provided.

Place: _____

Date: _____

Dr. PAPA RAO CHALLAGUNDLA